YieldLens; Crop yield prediction system for African farmers

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I. Introduction

Africa is the world's most rapidly growing continent both in demographics, economic and cultural growth. East and West Africa population alone is projected to grow to over 1.3 billion people (representing 60% of the African population by 2050). Despite the vast potential of human and land resources, African agriculture still faces severe constraints that threaten food security – unrealized potential in its uncultivated arable land, climate vulnerability, and high post-harvest losses **[[1]].** In 2020, over 280 million Africans representing a fifth of the population faced hunger, a further 346 million suffered from severe food insecurity and 452 million from moderate food insecurity. **[[2]]**

Traditional farming practices are inherently reliant on local resources, low external energy inputs and sometimes limited external inputs (water, soil, etc.). This makes them increasingly unreliable and vulnerable to ground factors like soil health depletion or changing weather. This reliance has led to low yields and wasted resources, while existing digital tools often lack integration, offline capabilities, and local data relevance for African contexts. [**[3]]**

While many solutions have been built to address the issue of food security, our proposed solution answers a very unique question, "***How do we make farmers make the right decision for post-harvest activities?*** This will enable farmers to prepare for harvest and adopt the right strategy for resource allocation and planning . This tool will also be powered with a multilingual AI chatbot that can advise the farmer not just before planting but also during planting.

II. Problem Statement

Most smallholder farmers in Africa lack access to reliable, data-driven tools that can help them anticipate how their crops will perform each planting season. Without timely yield estimates, farmers are left to rely on guesswork when making critical decisions around planting schedules, resource allocation and market preparation. The proposed system addresses this directly by leveraging climate data, Sentinel satellite imagery, and historical yield records to deliver accurate, season-by-season crop yield predictions. Alongside this, an AI-powered advisory interface enables farmers to ask questions, interpret their yield forecasts, and generate performance reports.

III. Proposed Solution

The crop yield prediction system is an integrated AI-powered platform designed to deliver personalized, sustainable crop recommendations based on climate data, geographical data, and historical crop yield.

The system follows a 4-layer architecture as shown in Fig 1.

IV. Aim and Objectives

General Objective To build and deploy an end-to-end AI-optimized crop yield prediction system to deliver personalized and sustainable crop yield predictions and agricultural advisory to farmers based

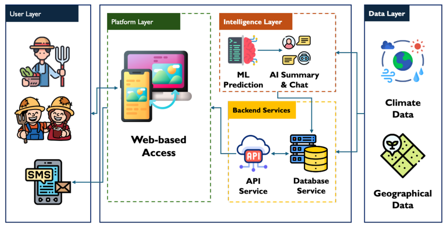

Fig. 1: user flow disgram

on climate data, geographical data and historical crop yield in Africa **Specific Objectives**

1) To develop an ML model that predicts crop yield based on soil properties (soil moisture), weather conditions, and climate data.
2) To develop an AI chatbot that provides personalised, interactive support before, during, and after the growing season.
3) To build and deploy a minimal but user-friendly platform accessible via web for large-scale farmers.
4) To build a 2G-based SMS network that delivers crop yield predictions directly to farmers in remote, low-connectivity areas without requiring smartphones or internet access.

V. Methodology

System Architecture

- **Frontend** – Mobile-first responsive web app (Flask + React) with offline capability.
- **Backend** – Flask (Python) with PostgreSQL database.
- **ML Pipeline** – scikit-learn, pandas, numpy for model training and inference.
- **Chatbot** – Integration of Gemini and OpenAI model APIs.
- **API Services** – RESTful APIs for data access and model predictions.

VI. Work Plan

A. List of Assumptions

1) The system assumes intermittent internet connectivity for the farmers.

Track	D1–D2 Setup	D3–D4 Build	D5–D6 Integrate	D7–D8 Finalise
Data & ML	Data collection & preprocessing	Model training & yield prediction	Refinement & advisory integration	Bug fixes & polish
GenAI Advisory	Scope prompts & advisory rules	Build & trigger LLM	Test on real predictions	QA & polish
Chatbot	Scope intents & conversation flow	Build chat & add web search	Ground in farmer data	QA & polish
Platform / Backend	Setup repo, API & DB schema	Build endpoints & wire prediction	Full integration & deployment	Testing & demo prep
PM / Docs / Deck	Research & project setup	Track progress	Track progress & capture outputs	Finalise report & deck

Fig. 2: Work plan

2) Most farmers in the region have access to basic smartphones.
3) We also assume a Sufficient number of farmers will participate to make collaborative learning effective.
4) We assume that the farmers will honestly engage in the yield data collection process.

B. List of Resources

1) Data Sources:

- **Kaggle Dataset for crop yield prediction** – Historical data of crop, country, year, season, area harvested, production, rainfall, yield.
- **Weather Data** – NOAA Weather.gov API, for temperature, precipitation, humidity.
- **Sentinel 2, DEM** – NDVI estimations.
- **CHIRPS** – Rainfall data.
- **ECMWF/ERAS_Land** – Temperature data.

VII. AI Use Disclosure

- Scopus AI was used to search through literature for the literature review.
- Scopus AI was used to search through literature for the methodology section.

References

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[2] T. Adebola and E. Ibeke, "Agriculture in africa: the emerging role of artificial intelligence," *LexisNexis Butterworths*, 2023.

[3] M. Agnoletti and A. Santoro, "Agricultural heritage systems and agrobiodiversity," *Biodiversity and Conservation*, vol. 31, no. 10, pp. 2231–2241, 2022.